QUESTION:
136. Single Number

Given a **non-empty** array of integers nums, every element appears *twice* except for one. Find that single one.
You must implement a solution with a linear runtime complexity and use only constant extra space.

Example 1:
Input: nums = [2,2,1]
Output: 1
Example 2:
Input: nums = [4,1,2,1,2]
Output: 4
Example 3:
Input: nums = [1]
Output: 1

Constraints:
- 1 <= nums.length <= 3 * 104
- -3 * 104 <= nums[i] <= 3 * 104
- Each element in the array appears twice except for one element which appears only once.

Hint:
There are three solutions for this one .

Most optimal one is :

As we know that xor operator “^” ; a^0=a. And a^a=0;

Using that we are going to solve . If a one element appears twice then it gives zero and atlas we get only one element that is the singly repeated elect in a;

Code:

```
Int singleelement(vector<inr > &arr){
        int a=0;
                for(auto i:arr){
                        a=a^i;
                        }
  Return  a;

}
```